

$\mathbb{Q}$ 是全序集

Lemma: $\forall x, y \in \mathbb{Q}$, 若 $x \geq 0$ 且 $y \geq 0$, 则有: $x + y \geq 0$

Proof: $\because x \in \mathbb{Q} \quad \therefore \exists m \in \mathbb{Z}, n \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } x = [m, n]$

$$\because x \geq 0 \quad \therefore mn \geq 0$$

$\because y \in \mathbb{Q} \quad \therefore \exists r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } y = [r, s]$

$$\because y \geq 0 \quad \therefore rs \geq 0$$

$$\therefore x + y = [m, n] + [r, s] = [ms + rn, ns]$$

$$\therefore (ms + rn)(ns) = (ms)(ns) + (rn)(ns) = (mn)s^2 + (rs)n^2 \geq 0$$

$$\therefore x + y = [ms + rn, ns] \geq 0 \quad \square$$

Lemma: $(\mathbb{Q}, \leq)$ 是全序集.

Proof: $\forall x \in \mathbb{Q}, \quad \because x - x = x + (-x) = 0 \geq 0 \quad \therefore x \geq x$

$$\therefore x \leq x \quad \therefore \text{反身性成立.}$$

$\forall x, y, z \in \mathbb{Q}$, 若 $x \leq y$ 且 $y \leq z$, 则有:

$$\because x \leq y \quad \therefore y \geq x \quad \therefore y - x \geq 0$$

$$\because y \leq z \quad \therefore z \geq y \quad \therefore z - y \geq 0 \quad \therefore (y - x) + (z - y) \geq 0$$

$$\therefore (y - x) + (z - y) = y + (-x) + z + (-y) = z + (-x) = z - x$$

$$\therefore z - x \geq 0 \quad \therefore z \geq x \quad \therefore x \leq z \quad \therefore \text{传递性成立.}$$

$\forall x, y \in \mathbb{Q}$, 若 $x \leq y$ 且 $y \leq x$, 则有:

$$\because x \leq y \quad \because y \geq x \quad \because y - x \geq 0 \quad \because -(y - x) \leq 0 \quad \because x - y \leq 0$$

$$\because y \leq x \quad \because x \geq y \quad \because x - y \geq 0$$

$$\because x - y \leq 0 \text{ 且 } x - y \geq 0 \quad \because x - y = 0 \quad \because x = y \quad \because \text{反称性成立.}$$

对 $\forall x, y \in \mathbb{Q}$,

若 $x = y$, 则有 $x \leq y$

若 $x < y$, 则有 $x \leq y$

若 $x > y$, 则有 $y < x$. $\therefore y \leq x$.

$\therefore x \leq y$ 或 $y \leq x$ 至少有一者成立.

$\therefore (\mathbb{Q}, \leq)$ 为全序集. $\square$

Lemma: $(\mathbb{Q}, \leq)$ 上的序结构限制在 $\mathbb{Z}$ 上给出 $(\mathbb{Z}, \leq)$ 的序结构.

Proof: 对 $\forall x, y \in \mathbb{Z}$.

若在 $\mathbb{Z}$ 中 $x \leq y$, 则有: $y - x \in \mathbb{Z}_{\geq 0}$. $\therefore y - x \geq 0 \quad \therefore y \geq x$

$\therefore x \leq y$ (在 $\mathbb{Q}$ 中)

若在 $\mathbb{Q}$ 中 $x \leq y$, 则有: $y \geq x$. $\therefore y - x \geq 0$.

$\therefore y - x \in \mathbb{Z}$ 且 $y - x \geq 0 \quad \therefore y - x \in \mathbb{Z}_{\geq 0} \quad \therefore x \leq y$ (在 $\mathbb{Z}$ 中)

$\square$

定义(有理数的分数记法) 对 $\forall [r, s] \in \mathbb{Q}$ (其中 $r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}$), 记 $\frac{r}{s} := [r, s] \in \mathbb{Q}$.

对 $\forall \frac{r_1}{s_1}, \frac{r_2}{s_2} \in \mathbb{Q}$, 有:

$$\frac{r_1}{s_1} + \frac{r_2}{s_2} = \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}$$

$$\frac{r_1}{s_1} \cdot \frac{r_2}{s_2} = \frac{r_1 r_2}{s_1 s_2}$$

定义(既约分式) 对 $\forall r \in \mathbb{Z}, \forall s \in \mathbb{Z} \setminus \{0\}$. 若 r 和 s 互素, 则称分式表达式 $\frac{r}{s}$ 是既约的.

定理: 每个有理数都有既约的分式表达式

Proof: 对于 $0 \in \mathbb{Q}$. $\because 0 \in \mathbb{Z}, 1 \in \mathbb{Z} \setminus \{0\}, \gcd(0, 1) = 1$

$\therefore 0 = \frac{0}{1}$ $\therefore \frac{0}{1}$ 是 $0 \in \mathbb{Q}$ 的既约分式表达式.

对于 $1 \in \mathbb{Q}$ $\because 1 \in \mathbb{Z}, 1 \in \mathbb{Z} \setminus \{0\}, \gcd(1, 1) = 1$

$\therefore 1 = \frac{1}{1}$ $\therefore \frac{1}{1}$ 是 $1 \in \mathbb{Q}$ 的既约分式表达式.

对于 $-1 \in \mathbb{Q}$ $\because -1 \in \mathbb{Z}, 1 \in \mathbb{Z} \setminus \{0\}, \gcd(-1, 1) = 1$

$\therefore -1 = \frac{-1}{1}$ $\therefore \frac{-1}{1}$ 是 $-1 \in \mathbb{Q}$ 的既约分式表达式.

对 $\forall \frac{m}{n} \in \mathbb{Q}$ (其中 $m \in \mathbb{Z}, n \in \mathbb{Z} \setminus \{0\}$) 将 m 和 n 分解成素因子的乘积:

$n = \pm \prod_{i=1}^r p_i^{a_i}$, $m = \pm \prod_{i=1}^r p_i^{b_i}$, 其中 $r \in \mathbb{Z}_{\geq 0}$, $p_1, \dots, p_r$ 是相异素数,

$a_i, b_i \in \mathbb{Z}_{\geq 0}$.

则有:
$$\frac{m}{n} = \frac{\pm \prod_{i=1}^r p_i^{b_i}}{\pm \prod_{i=1}^r p_i^{a_i}} = \pm \frac{\prod_{i=1}^r p_i^{\min\{a_i, b_i\}} \cdot \prod_{i=1}^r p_i^{b_i - \min\{a_i, b_i\}}}{\prod_{i=1}^r p_i^{\min\{a_i, b_i\}} \cdot \prod_{i=1}^r p_i^{a_i - \min\{a_i, b_i\}}}$$

$$= \pm \frac{\prod_{i=1}^r p_i^{b_i - \min\{a_i, b_i\}}}{\prod_{i=1}^r p_i^{a_i - \min\{a_i, b_i\}}}$$

是既约的分式表达式. □

Lemma: $\frac{r_1}{s_1}, \frac{r_2}{s_2} \in \mathbb{Q}$, 且 $\frac{r_1}{s_1}$ 和 $\frac{r_2}{s_2}$ 是既约的分式表达式. 则有:

$$\frac{r_1}{s_1} = \frac{r_2}{s_2} \Leftrightarrow (r_1 = r_2 \text{ 且 } s_1 = s_2) \text{ 或 } (r_1 = -r_2 \text{ 且 } s_1 = -s_2)$$

Proof: $\frac{r_1}{s_1} = \frac{r_2}{s_2} \Leftrightarrow r_1 s_2 = r_2 s_1$

$$\begin{aligned} (\Rightarrow): \because r_1 = \frac{r_2 s_1}{s_2} \in \mathbb{Z} \quad \therefore s_2 \mid r_2 s_1 \quad \because \gcd(r_2, s_2) = 1 \quad \therefore s_2 \mid s_1 \\ \therefore r_2 = \frac{r_1 s_2}{s_1} \in \mathbb{Z} \quad \therefore s_1 \mid r_1 s_2 \quad \because \gcd(s_1, r_1) = 1 \quad \therefore s_1 \mid s_2 \end{aligned}$$

$$\therefore s_1 = \pm s_2$$

若 $s_1 = s_2$, 则 $r_1 = r_2$. 若 $s_1 = -s_2$, 则 $r_1 = -r_2$

$$\therefore (r_1 = r_2 \text{ 且 } s_1 = s_2) \text{ 或 } (r_1 = -r_2 \text{ 且 } s_1 = -s_2)$$

($\Leftarrow$): 显然. □